

Machine Learning-IS51

Unit I

Introduction: Need for Machine Learning, Machine Learning Explained, Machine Learning in Relation to other Fields, Types of Machine Learning, Challenges of Machine Learning, Machine Learning Process, Machine Learning Applications.

Understanding Data – 1: Introduction, Big Data Analysis Framework, Descriptive Statistics, Univariate Data Analysis and Visualization, Bivariate Data and Multivariate Data, Multivariate Statistics

Text Book:

1. S Sridhar, M Vijayalakshmi, “Machine Learning”, OXFORD University Press 2021, First Edition.
2. T. M. Mitchell, “Machine Learning”, McGraw Hill, 1997.

Reference Books:

1. M N Murty and Ananthanarayana V S, “Machine Learning: Theory and Practice”, Universities Press (India) Pvt. Limited, 2024.
2. Stephen Marsland, “Machine Learning - An Algorithmic Perspective”, Second Edition, CRC Press - Taylor and Francis Group, 2015.
3. Jiawei Han, Micheline Kamber and Jian Pei, Data Mining: Concepts and Techniques, 3rd ed, The Morgan Kaufmann Series in Data Management Systems Morgan Kaufmann Publishers, July 2011.

Topics to be covered

Introduction:

- 1.1 Need for Machine Learning
- 1.2 Machine Learning Explained
- 1.3 Machine Learning in Relation to other Fields
- 1.4 Types of Machine Learning
- 1.5 Challenges of Machine Learning
- 1.6 Machine Learning Process
- 1.7 Machine Learning Applications

Understanding Data – 1: Introduction

- 2.3 Big Data Analysis Framework
- 2.4 Descriptive Statistics
- 2.5 Univariate Data Analysis and Visualization
- 2.6 Bivariate Data and Multivariate Data
- 2.7 Multivariate Statistics

1.1 Need for Machine Learning

- Business organizations generate large amounts of data every day.
- Earlier, this data was not fully used because it was stored in different systems and was difficult to combine.
- There was also a lack of advanced software tools to analyze the data and find useful information.
- Today, organizations use modern technologies such as Machine Learning to analyze data and make better decisions.

Why is Machine Learning Becoming Popular?

Machine Learning (ML) has become popular mainly because of three major reasons:

1. Huge Amount of Data

- Companies such as Facebook, Twitter, and YouTube generate massive amounts of data.
- Data is growing very rapidly, approximately doubling every year.
- ML helps computers analyze this large amount of data.

2. Lower Storage and Hardware Cost

- The cost of storing data has decreased significantly.
- Computer hardware has also become cheaper and more powerful.
- Therefore, organizations can easily collect, store, process, and transmit data.

3. Availability of Advanced Algorithms

- Many powerful ML algorithms are now available.
- Deep Learning has further increased the capabilities of ML.
- These algorithms can solve complex real-world problems.

The Knowledge Pyramid explains how raw data becomes useful for decision-making.

1.1 Cont'd

Data → Information → Knowledge → Intelligence → Wisdom

❑ **Data:** Raw facts, numbers, symbols, or text.

Example: **100**, **150**, **200** sales records.

❑ **Information:** Processed data that reveals patterns or relationships.

Example: Product A has the highest sales.

❑ **Knowledge:** Useful conclusions obtained from information.

Example: Product A has been consistently popular over the last year.

❑ **Intelligence:** Applying knowledge to take appropriate action.

Example: Increase production of Product A.

❑ **Wisdom:** Ability to make mature and effective decisions based on experience and understanding.

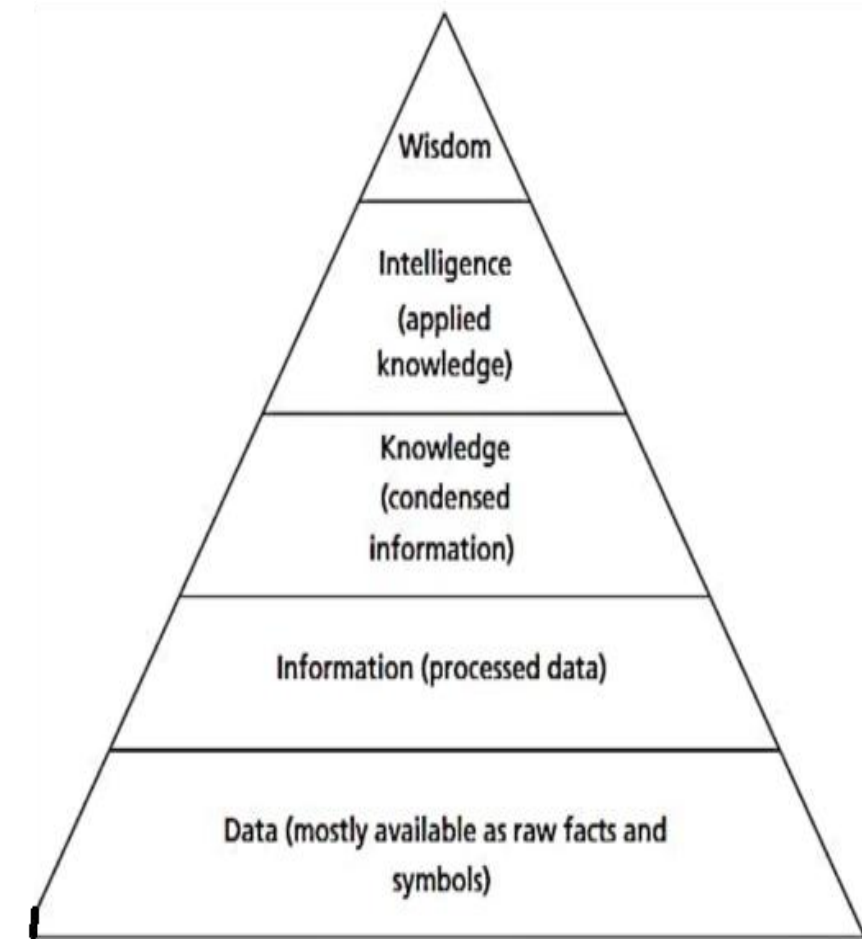


Figure 1.1: The Knowledge Pyramid

Why Do We Need Machine Learning?

Organizations collect large amounts of historical data.

Machine Learning helps to:

- Analyze large datasets
- Identify hidden patterns and trends
- Make better decisions
- Predict future outcomes
- Design new products
- Improve business processes
- Develop effective Decision Support Systems

Machine Learning converts large amounts of data into Machine Learning converts large amounts of data into useful insights that help organizations make better decisions.

Simple example:

Sales Data → Sales Pattern → Product Knowledge → Production Decision → Better Business Outcome

1.2 Machine Learning Explained

ML is a part of AI. Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

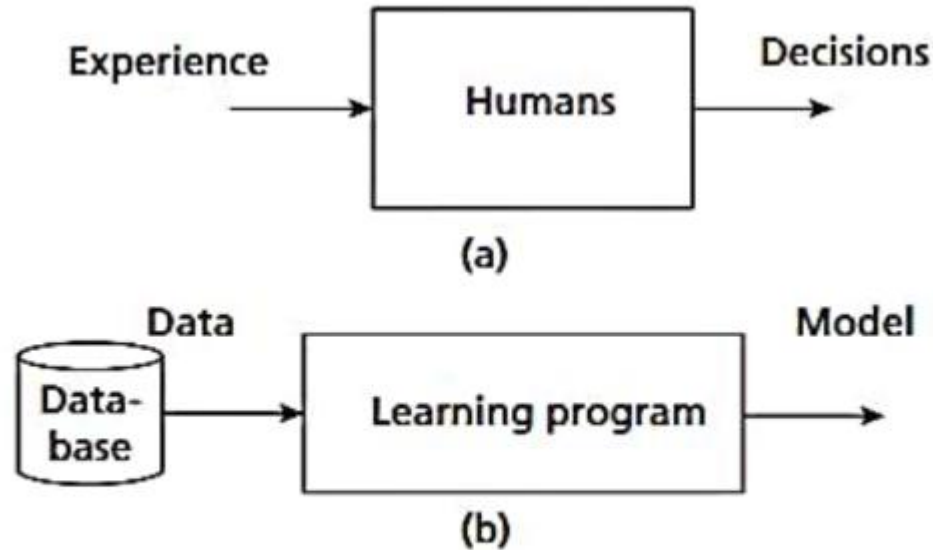


Figure 1.2: (a) A Learning System for Humans (b) A Learning System for Machine Learning

1.2 Cont'd

Conventional Programming

- Rules are manually written
- Programmer defines the logic
- Input + Rules $\rightarrow$ Output
- Difficult for highly complex problems
- Does not automatically learn
- Example: Calculator program

Machine Learning

- Patterns are learned from data
- Algorithm learns the logic
- Data + Examples $\rightarrow$ Model
- Suitable for complex data-driven problems
- Can learn and improve from data
- Example: Spam detection

Limitations of Conventional Programming

Conventional programming works well when rules are clearly defined.

However, it becomes difficult for complex real-world problems such as:

- **Puzzles**
 - **Games**
 - **Image Recognition**
 - **Natural Language Processing**
 - **Autonomous systems**
-
- It is difficult to manually write rules for every possible situation.
 - Complex problems may have too many possible conditions.
 - Human-designed rules may not handle unknown situations effectively.

Early Artificial Intelligence – Rule-Based Approach

The early approach of AI was to create general-purpose rules manually.

1. Understand the problem.
2. Obtain knowledge from human experts.
3. Convert expert knowledge into rules.
4. Represent rules using logic and reasoning.
5. Implement the rules in a computer program.
6. Use the program to make decisions.

Expert Knowledge → Rules → Program → Decision

Expert Systems

An Expert System is an AI system that uses the knowledge of human experts to solve problems in a specific domain.

Main components:

- Knowledge Base – stores expert knowledge and rules.
- Inference Engine – applies rules to available information.
- User Interface – allows users to interact with the system.

Example: MYCIN

- MYCIN was an early expert system for medical diagnosis.
- It used knowledge obtained from doctors.
- The system applied predefined rules to assist in diagnosing infections and recommending antibiotics.

Limitations of Expert Systems

Although expert systems were useful, they had several limitations:

- They depended heavily on human experts.
- Rules had to be manually created and updated.
- Difficult to handle new or unexpected situations.
- Difficult to scale when the number of rules becomes very large.
- They did not truly learn from data.
- Hence, they had limitations in demonstrating genuine intelligence.

Human must provide the rules → System follows the rules

Shift Towards Machine Learning

Because rule-based systems had limitations, AI gradually shifted toward a **data-driven approach**.

Instead of manually writing all the rules:

- Collect relevant data.
- Give the data to a machine learning algorithm.
- The algorithm identifies patterns and relationships.
- A model is automatically developed.
- The model can make predictions on new/unseen data.

Data → Learning Algorithm → Model → Prediction

Machine Learning (ML) has become popular mainly because of **three major reasons**:

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- Data is growing very rapidly, approximately **doubling every year**.
- ML helps computers analyze this large amount of data.

2. Lower Storage and Hardware Cost

- The cost of storing data has decreased significantly.
- Computer hardware has also become cheaper and more powerful.
- Therefore, organizations can easily **collect, store, process, and transmit data**.

3. Availability of Advanced Algorithms

- Many powerful ML algorithms are now available.
- Deep Learning has further increased the capabilities of ML.
- These algorithms can solve complex real-world problems.

1. Data Collection

- First, relevant data is collected.
- The quality and quantity of data are important for learning.

2. Abstraction

- The collected data is analyzed to identify important concepts and patterns.
- These concepts help the system understand objects or situations.
- Example: From many images of elephants, the system learns the concept of what an elephant looks like.

3. Generalization

- The learned concepts are converted into useful knowledge for making decisions.
- The system identifies patterns, ranks possible concepts, and develops heuristics.
- Heuristics are practical rules of thumb learned from experience.
- Example: From past experiences, a person learns to recognize situations that may indicate danger.

4. Evaluation and Correction

- Learned rules or heuristics may sometimes produce incorrect results.
- Evaluation measures are used to check how well the model performs.
- If performance is poor, the model is corrected or improved.

Data Collection → Abstraction → Generalization → Evaluation → Improved Model

1.3 Machine learning in relation to other field

- Machine learning is an important branch of AI.
- Machine learning is the subbranch of AI, whose aim is to extract the patterns for prediction.
- The aim of AI is to develop intelligent agents. An agent can be a robot, humans, or any autonomous systems.
- Deep learning is a subbranch of machine learning. In deep learning, the models are constructed using neural network technology. Many neurons form a network connected with the activation functions that trigger further neurons to perform tasks.
- Machine learning starts with data. Therefore, data science and machine learning are interlinked.

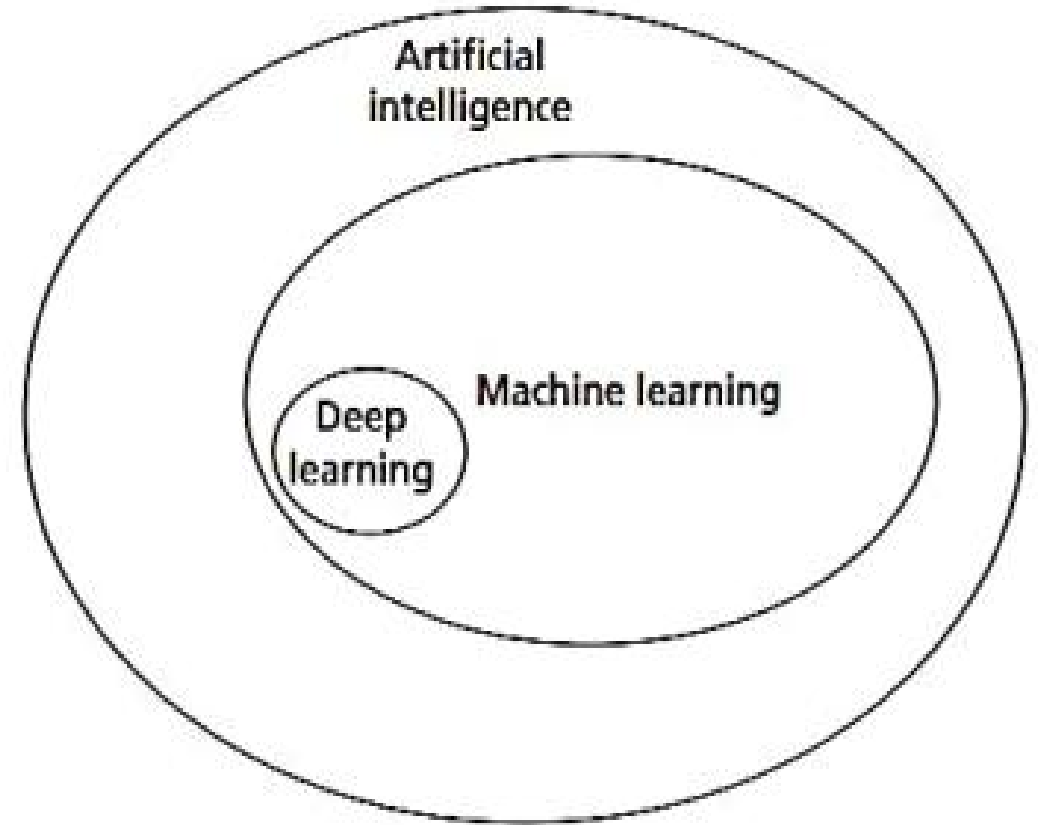


Figure 1.3: Relationship of AI with Machine Learning

Data science deals with gathering of data for analysis It is a broad field that includes:

1. Big Data

Big Data is a specialization within data science focused on managing data with three primary characteristics:

- **Volume:** Large quantities of data created by major tech platforms (e.g., social media and video sites).
- **Variety:** Diverse formats and media types, including images and video files.
- **Velocity:** The rapid rate at which information is generated and processed.

It powers machine learning tasks like translation and image recognition, and drives advancements in deep learning (which builds models using neural networks).

2. Data Mining

Originating from business needs, data mining focuses on uncovering hidden insights within datasets that might otherwise be overlooked. While often conflated with machine learning, they differ in purpose:

- **Data Mining:** Focused on discovering hidden patterns in existing data.
- **Machine Learning:** Focused on using data patterns to make predictions.

3. Data Analytics

Data analytics transforms raw data into actionable knowledge. A key subtype, **Predictive Analytics**, relies on machine learning algorithms to forecast future outcomes.

4. Pattern Recognition

An engineering domain that uses machine learning algorithms to identify features, analyze structures, and classify patterns. It can be understood as a specific practical application of machine learning.

1.3 Cont'd

These relations are summarized in Figure 1.4.

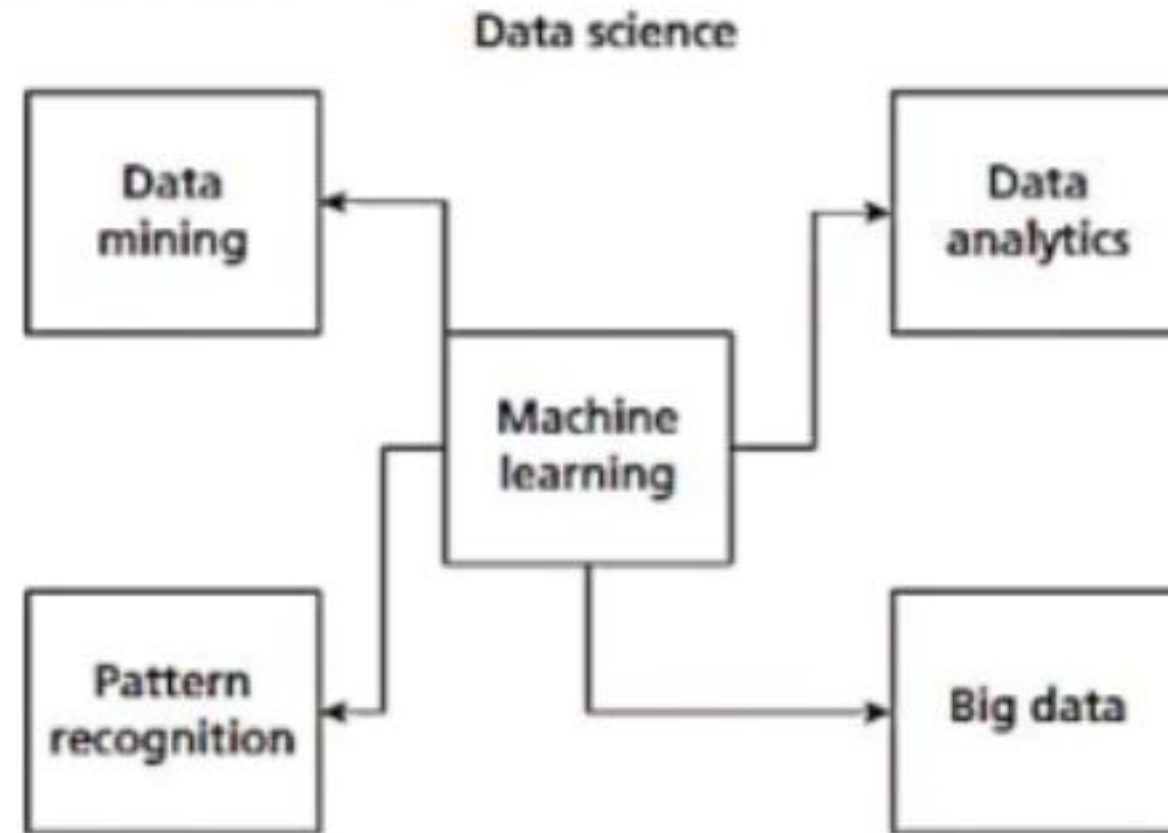


Figure 1.4: Relationship of Machine Learning with Other Major Fields

Statistics:

- A mathematical branch with strong, rigorous theoretical foundations. It relies on hypothesis testing and experiments to validate data relationships.
- Math-heavy, using complex equations and strict assumptions tailored to specific datasets. Requires deep statistical knowledge.

Machine Learning:

- A computational approach focused on learning directly from data with fewer underlying assumptions.
- Requires less formal statistical background but relies heavily on software tools to automate the learning process.

“Many views consider **Machine Learning** as the modern, automated evolution of "old Statistics".

1.4 Types of Machine Learning

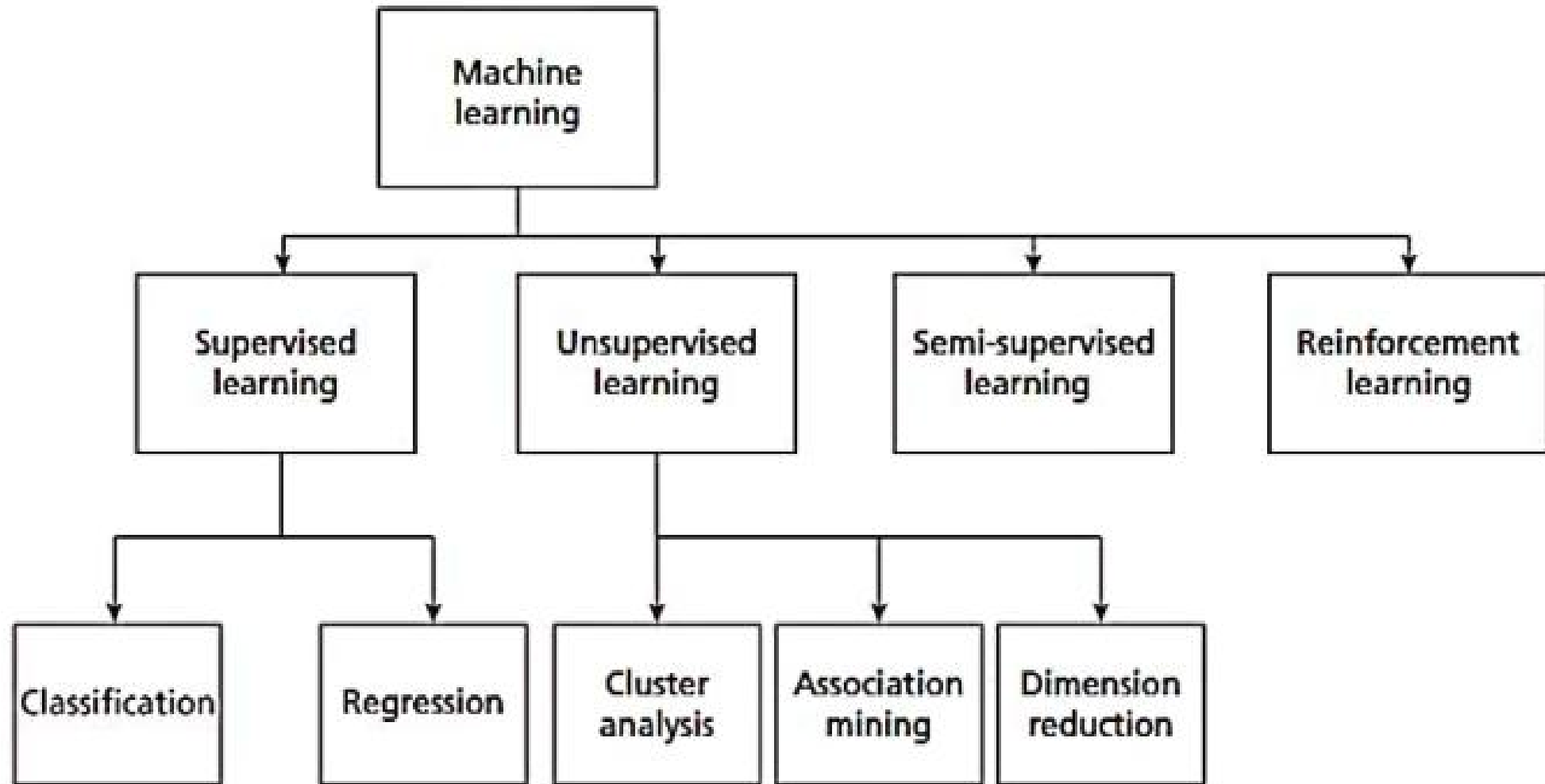
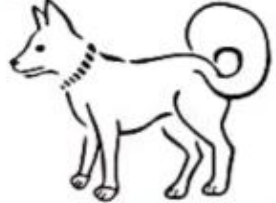


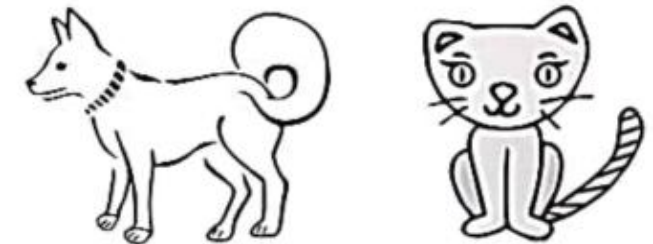
Figure 1.5: Types of Machine Learning

Dataset

S.No.	Length of Petal	Width of Petal	Length of Sepal	Width of Sepal	Class
1.	5.5	4.2	1.4	0.2	Setosa
2.	7	3.2	4.7	1.4	Versicolor
3.	7.3	2.9	6.3	1.8	Virginica

Input	Label
	dog
	Cat

(a)



(b)

Labelled and Unlabelled Data

Figure 1.6: (a) Labelled Dataset (b) Unlabelled Dataset

Supervised learning:

classification :

Some of the key algorithms of classification are:

- Decision Tree
- Random Forest
- Support Vector Machines
- Naïve Bayes
- Artificial Neural Network and Deep Learning networks like CNN

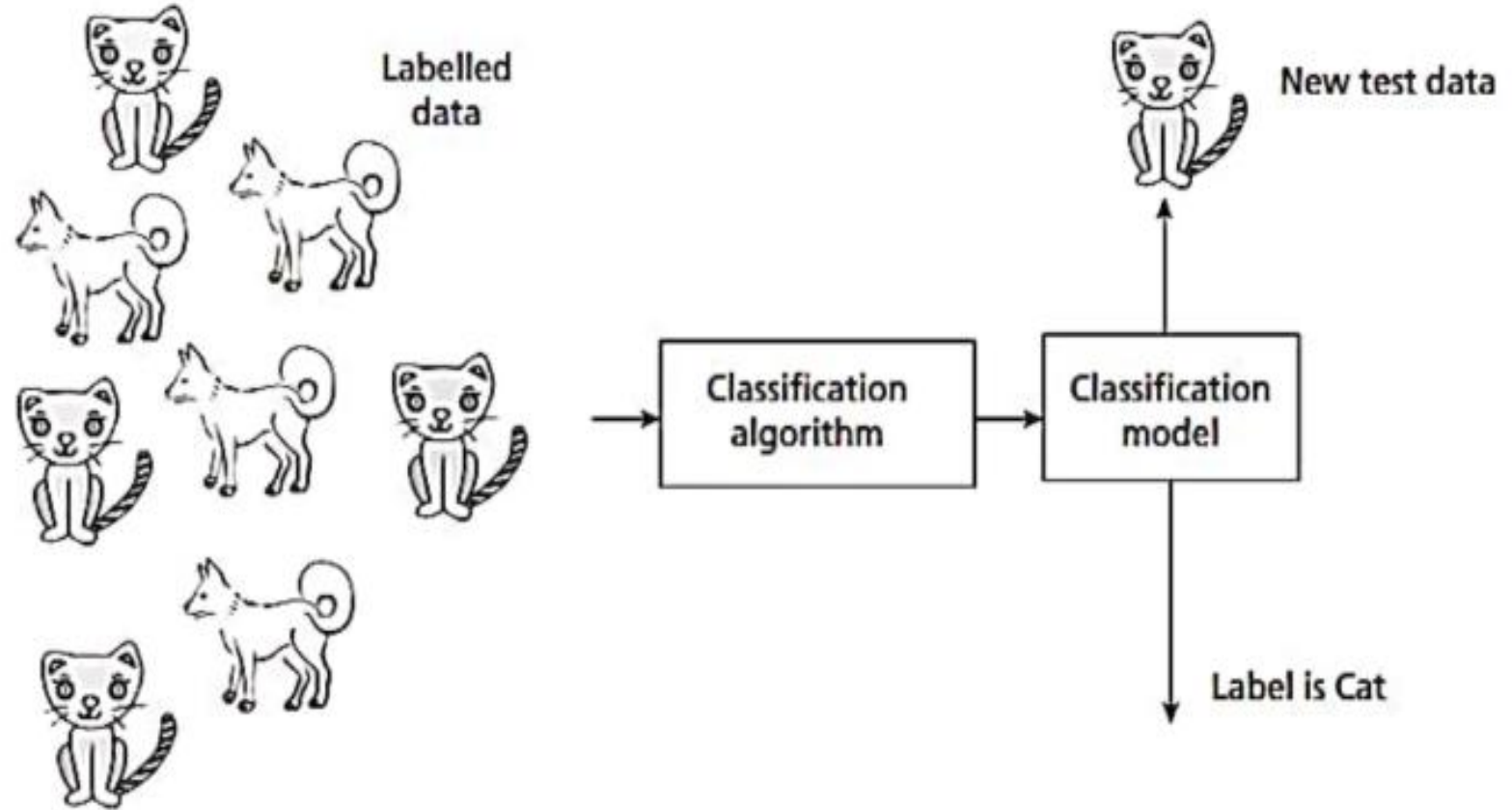


Figure 1.7: An Example Classification System

Regression Models

Regression models predict continuous variables (such as product sales or product price).

The regression model takes input x and generates a model in the form of a fitted line of the form $y = f(x)$.

Here, x is the independent variable that may be one or more attributes and y is the dependent variable.

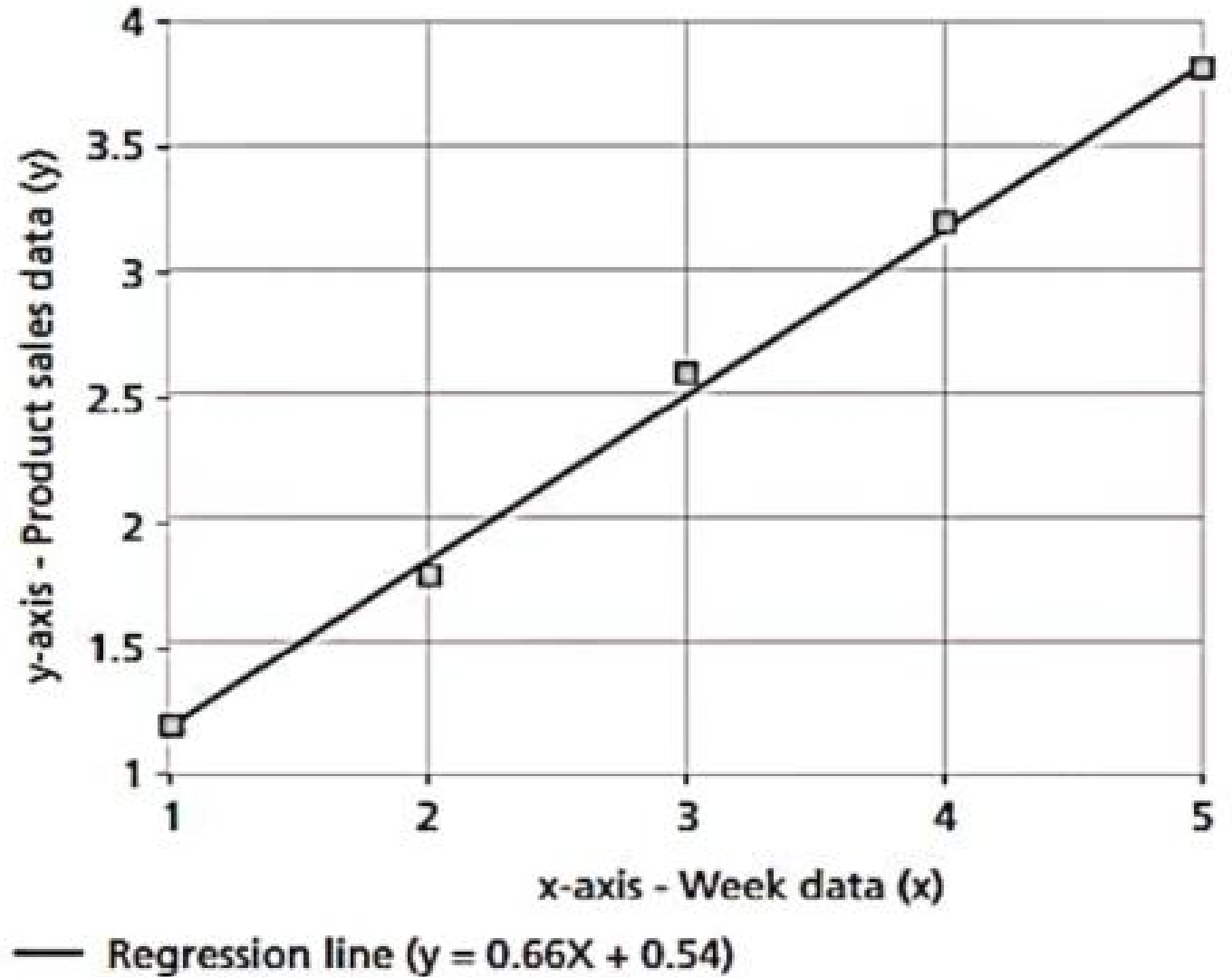


Figure 1.8: A Regression Model of the Form $y = ax + b$

Comparison

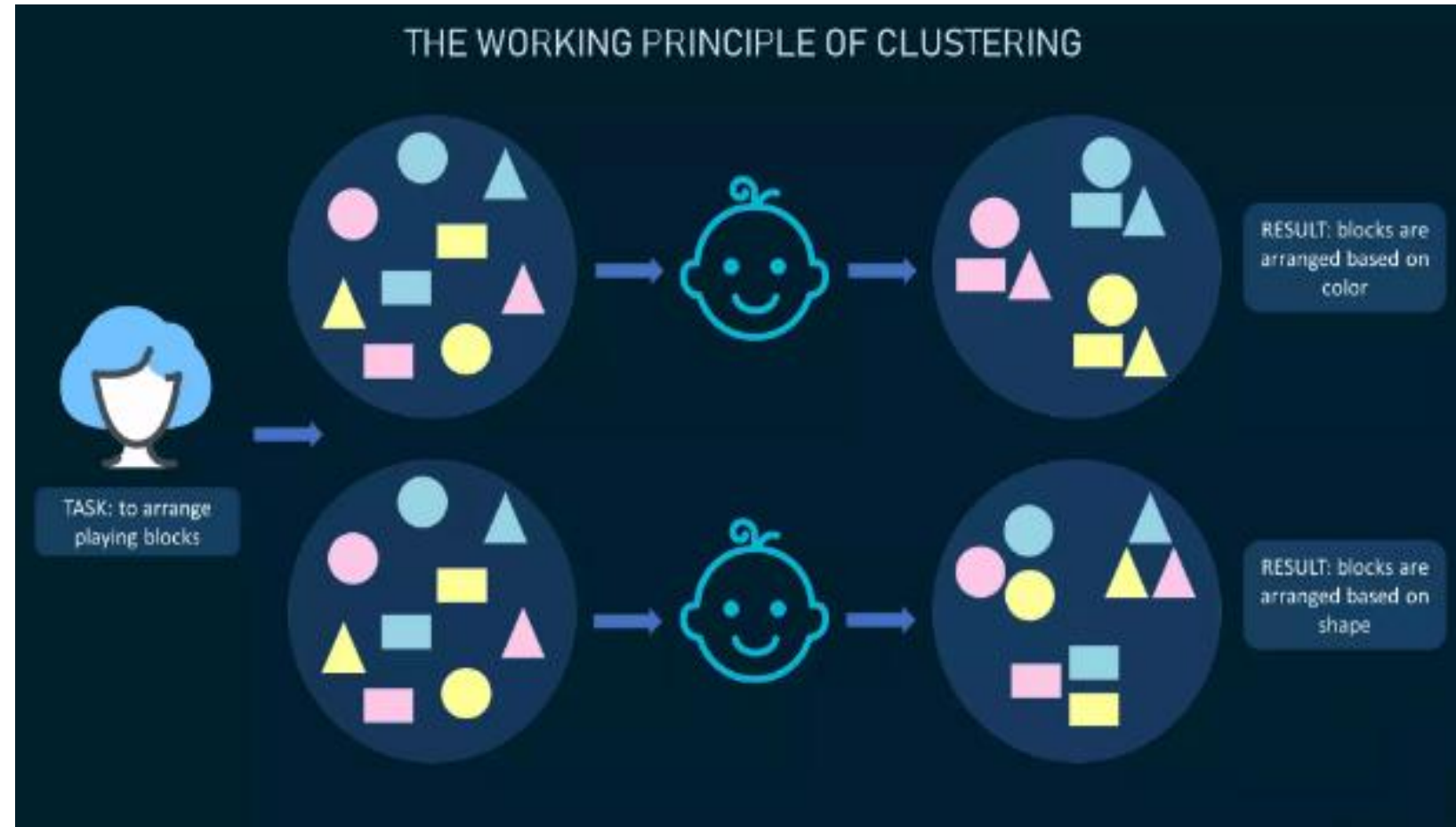
Classification Models	Regression Models
Categorical / Discrete	Continuous Numerical
Find decision boundaries to separate classes	Fit a line/curve to capture trends ($y = ax + b$)
Accuracy, F1-Score, ROC-AUC	MSE, RMSE, MAE, R^2 Score
Cross-Entropy Loss	Mean Squared Error (L2)
Spam vs. Non-Spam Detection	Housing Price / Product Sales Prediction
Decision Trees, Random Forest, Support Vector Machines, Naïve Bayes, Neural Networks / CNNs	Linear Regression, Polynomial Regression

Unsupervised Learning

A machine learning model tries to find any similarities, differences, patterns, and structure in data by itself. No prior human intervention is needed.

Example: Anomaly detection, Customer and market segmentation, recommender system

- Clustering
- Association



Unsupervised Learning

- Operates without a supervisor, teacher, or target labels.
- Relies on finding inherent patterns, structures, and grouping principles directly from unlabelled data.
- Key algorithmic families include **Cluster Analysis and Dimensionality Reduction**.

Applications:

- **Image Processing:** Segmentation of Regions of Interest (ROI) in images.
- **Medical Imaging:** Automated detection of abnormal growths or tumors.
- **Bioinformatics:** Pattern matching and signature clustering in gene databases.
- **Computer Vision Example:** Grouping mixed image sets (e.g., separating images of cats and dogs) into distinct clusters based purely on visual attributes.

Cluster Analysis :

- Cluster analysis is an unsupervised learning technique that groups unlabelled data into disjoint clusters based purely on feature attributes.
- Ensures high intra-cluster similarity (objects within a group share common traits) and high inter-cluster variation (objects across groups differ significantly).
- Ensures high intra-cluster similarity (objects within a group share common traits) and high inter-cluster variation (objects across groups differ significantly).
- Automatically separates the dataset into distinct clusters (e.g., Cluster 1: Cats, Cluster 2: Dogs).

Some of the clustering algorithms are

k-means algorithm:

A centroid-based partition technique that iteratively assigns data points to k pre-specified clusters by minimizing within-cluster variance (Euclidean distance).

Hierarchical algorithms: Structure-based approaches that build a nested tree of clusters (dendrogram) using either a **bottom-up (agglomerative)** or **top-down (divisive)** strategy.

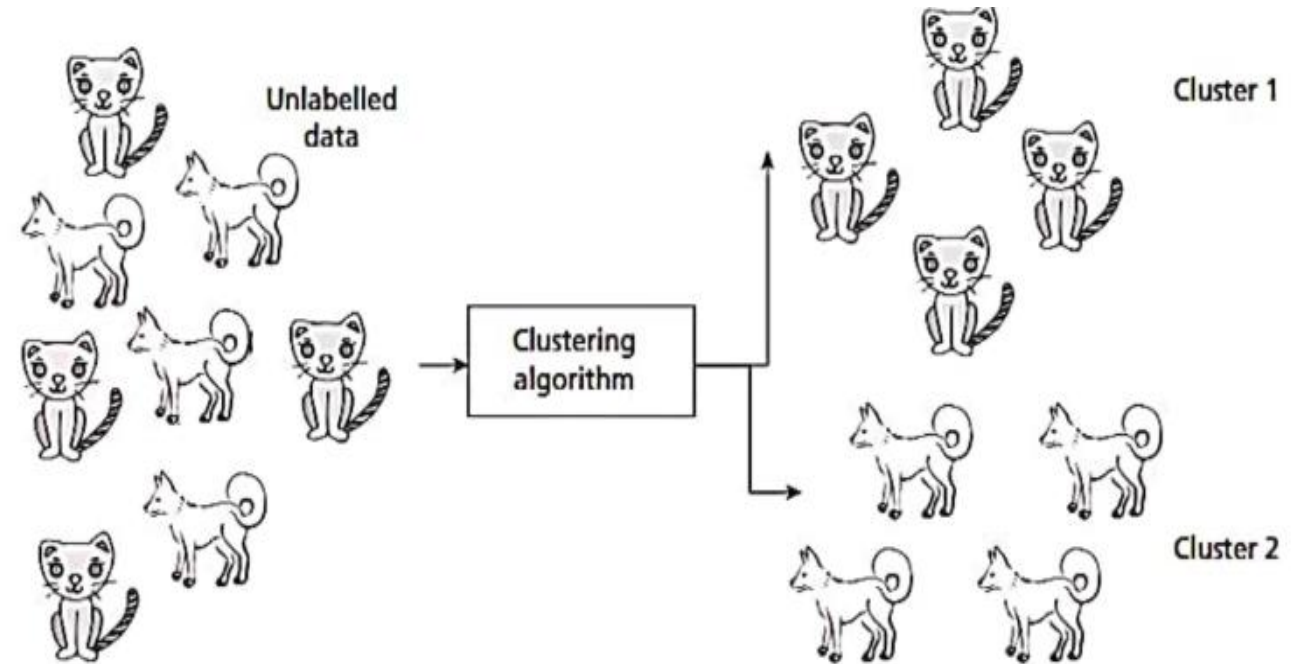
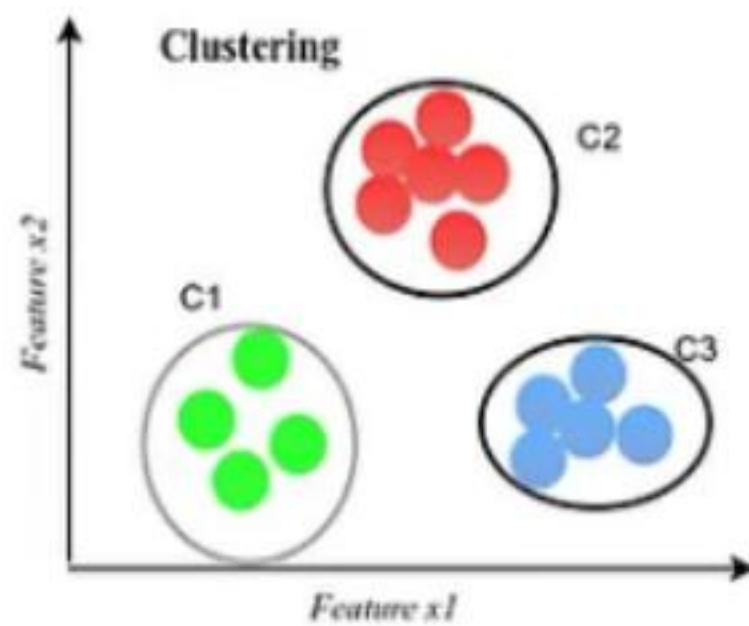
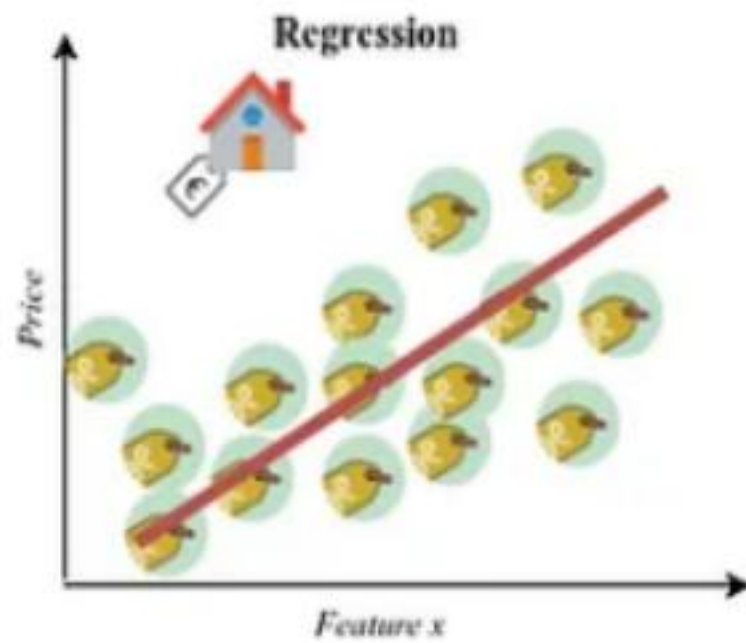


Figure 1.9: An Example Clustering Scheme

Dimensionality Reduction

- Dimensionality reduction algorithms fall under unsupervised learning.
- Accepts high-dimensional data as input and transforms it into a lower-dimensional space.
- Shrinks the dataset feature count while preserving its overall structure and generality (mitigating the "curse of dimensionality").
- Reducing input dimensions significantly lowers training time, memory consumption, and algorithmic complexity.
- Noise removal before feeding data into predictive models.
- Common Techniques includes Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA).



MEASURING MACHINE
LEARNING MODELS
SUPERVISED
LEARNING

Classification Metrics

Metric	Formula	Meaning	Better Value
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Percentage of correctly classified samples	Higher
Precision	$\frac{TP}{TP+FP}$	Of all predicted positives, how many are actually positive	Higher
Recall / Sensitivity	$\frac{TP}{TP+FN}$	Of all actual positives, how many are correctly detected	Higher
Specificity	$\frac{TN}{TN+FP}$	Of all actual negatives, how many are correctly identified	Higher
F1-score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Balance between Precision and Recall	Higher
ROC-AUC	Area under ROC curve	Ability to distinguish between classes across thresholds	Higher

What is Confusion matrix?

A confusion matrix is an $N \times N$ matrix used for evaluating the performance of a classification model, where N is the number of target classes.

This compares the actual target values with predicted target values.

		Actual Values	
		Positive	Negative
Predicted Values	Positive	TP	FP
	Negative	FN	TN

confusion matrix for binary classification problems

n=200	Predicted:		
	No	Yes	
Actual: No	TN=100	FP=15	115
Actual: Yes	FN=5	TP=80	85
	105	95	

True Negatives (TN): The patients actually don't have disease and our model also says that these patients don't have disease

True Positives (TP): These are cases in which, The patients actually have disease and our model also says that these patients do have disease

False Positives (FP): These are cases in which, The patients actually don't have disease but our model says that these patients do have disease

False Negatives (FN): These are cases in which, The patients actually have disease but our model says that these patients don't have disease

Performance measures

ACCURACY

The no of predictions that the model got right

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) = 80 + 100 / 200 = 0.9$$

Error Rate

The no of predictions that the model predicted wrongly

$$\text{Error Rate} = (FP + FN) / (TP + TN + FP + FN) = 15 + 5 / 200 = 0.1$$

Precision

When the model predicts the positives, how often is it right?

$$\text{Precision} = TP / (TP + FP) = 80 / 95 = 0.86$$

Recall

When it's actually yes, how often does it predict yes?

$$\text{Recall} = (TP) / (TP + FN) = 80 / 85 = 0.94$$

	Predicted: No	Predicted: Yes	
Actual: No	TN=100	FP=15	115
Actual: Yes	FN=5	TP=80	85
	105	95	

Example

- **Accuracy:** Overall, how often is the classifier correct?
 - $(TP+TN)/total = (100+50)/165 = 0.91$
- **Misclassification Rate:** Overall, how often is it wrong?
 - $(FP+FN)/total = (10+5)/165 = 0.09$
 - equivalent to 1 minus Accuracy
 - also known as "Error Rate"
- **True Positive Rate:** When it's actually yes, how often does it predict yes?
 - $TP/actual\ yes = 100/105 = 0.95$
 - also known as "Sensitivity" or "Recall"
- **False Positive Rate:** When it's actually no, how often does it predict yes?
 - $FP/actual\ no = 10/60 = 0.17$
- **True Negative Rate:** When it's actually no, how often does it predict no?
 - $TN/actual\ no = 50/60 = 0.83$
 - equivalent to 1 minus False Positive Rate
 - also known as "Specificity"
- **Precision:** When it predicts yes, how often is it correct?
 - $TP/predicted\ yes = 100/110 = 0.91$
- **Prevalence:** How often does the yes condition actually occur in our sample?
 - $actual\ yes/total = 105/165 = 0.64$

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

confusion matrix for multiclass classification

True Class	Predicted Class		
	1	2	3
1	8	2	0
2	1	9	0
3	1	2	7

True Class	Predicted Class	
	1 – Yes	2 & 3 – No
1 – Yes	TP - 8	FN - 2
2 & 3 – No	FP - 2	TN - 18

True Class	Predicted Class	
	2 – Yes	1 & 3 – No
2 – Yes	TP - 9	FN - 1
1 & 3 – No	FP - 4	TN - 16

True Class	Predicted Class	
	3 – Yes	1 & 2 – No
3 – Yes	TP - 7	FN - 3
1 & 2 – No	FP - 0	TN - 20

- $Accuracy(class - 1) = \frac{TP+TN}{TP+FN+FP+TN}$
- $Accuracy(class - 1) = \frac{8+18}{8+2+2+18} = \frac{26}{30} = \underline{0.87}$
- $Accuracy(class - 2) = \frac{TP+TN}{TP+FN+FP+TN}$
- $Accuracy(class - 2) = \frac{9+16}{9+1+4+16} = \frac{25}{30} = \underline{0.83}$
- $Accuracy(class - 3) = \frac{TP+TN}{TP+FN+FP+TN}$
- $Accuracy(class - 3) = \frac{7+20}{7+3+0+20} = \frac{27}{30} = 0.90$

- $Precision(class - 1) = \frac{TP}{TP+FP}$
- $Precision(class - 1) = \frac{8}{8+2} = \frac{8}{10} = 0.8$
- $Recall(class - 1) = \frac{TP}{TP+FN}$
- $Recall(class - 1) = \frac{8}{8+2} = \frac{8}{10} = 0.8$
- $F - 1 Score(class - 1) = \frac{2*Precision*Recall}{Precision+Recall}$
- $F - 1 Score(class - 1) = \frac{2*0.8*0.8}{0.8+0.8} = \frac{1.28}{1.6} = 0.8$

- $Precision(class - 2) = \frac{TP}{TP+FP}$
- $Precision(class - 2) = \frac{9}{9+4} = \frac{9}{13} = 0.69$
- $Recall(class - 2) = \frac{TP}{TP+FN}$
- $Recall(class - 2) = \frac{9}{9+1} = \frac{9}{10} = 0.9$
- $F - 1\ Score(class - 2) = \frac{2*Precision*Recall}{Precision+Recall}$
- $F - 1\ Score(class - 2) = \frac{2*0.69*0.9}{0.69+0.9} = \frac{1.242}{1.59} = 0.78$

- $Precision(class - 3) = \frac{TP}{TP+FP}$
- $Precision(class - 3) = \frac{7}{7+0} = \frac{7}{7} = 1.0$
- $Recall(class - 3) = \frac{TP}{TP+FN}$
- $Recall(class - 3) = \frac{7}{7+3} = \frac{7}{10} = 0.7$
- $F - 1\ Score(class - 3) = \frac{2*Precision*Recall}{Precision+Recall}$
- $F - 1\ Score(class - 3) = \frac{2*1.0*0.7}{1.0+0.7} = \frac{1.242}{1.59} = 0.82$

- *Macro F1 – Score* =

$$\frac{F1\text{-score}(C-1)+F1\text{-score}(C-2)+F1\text{-score}(C-3)}{3}$$

- *Macro F1 – Score* = $\frac{0.8+0.78+0.82}{3} = 0.8$

- *Weighted F1* = $\frac{10*0.8+10*0.78+10*0.82}{10+10+10} = 0.8$

weighed F1=

(number of class-1 samples(i.e 10)*F1-score(C-1)+
 number of class-2 samples(i.e 10)*F1-score(C-2)+
 number of class-3 samples(i.e 10)*F1-score(C-3)) / total number of samples of all
 classes(i.e 10+10+10)

1.5 Challenges of Machine Learning

Computers are better than humans at fast computation and repetitive tasks, and they can perform complex activities such as chess and Go. Humans, however, are generally better at recognition and understanding tasks. Deep learning has reduced this gap by enabling machines to recognize faces and patterns efficiently. Nevertheless, machine learning systems depend heavily on large amounts of high-quality data, making data quality and availability major challenges in building effective models.

1. Problems

- ◆ Machine learning works well with well-posed problems, where the problem specifications and required information are clearly defined.
- ◆ Ill-posed problems occur when the available data is insufficient to determine a unique solution.
- ◆ In the given example, the same data can be explained by different functions such as multiplication, division, or squaring.
- ◆ Therefore, additional examples or data are needed to identify the correct model.
- ◆ Puzzles, games, scientific computing, and real-world problems can become ill-posed when their specifications or information are incomplete.

Table 1.3: An Example

Input (x_1, x_2)	Output (y)
1, 1	1
2, 1	2
3, 1	3
4, 1	4
5, 1	5

2. Huge Data

- Machine learning requires large amounts of high-quality data.
- Data should be accurate and free from problems such as missing or incorrect values.
- Collecting and maintaining quality data is a major challenge.

3. High Computing Power

- Big Data and complex ML tasks require high computational resources.
- GPUs and TPUs are often used to speed up machine learning computations.
- Increasing algorithm complexity also increases time and computational requirements.

4. Complexity of Algorithms

- Choosing, designing, implementing, and evaluating suitable ML algorithms can be challenging.
- Different algorithms may need to be compared to identify an appropriate solution.
- Algorithm selection and optimization are important challenges for ML professionals.

5. Bias–Variance Tradeoff

- Bias and variance are important sources of model error.
- Overfitting: The model performs very well on training data but poorly on test data, resulting in poor generalization.
- Underfitting: The model performs poorly even on training data but may generalize reasonably well.
- Finding the right balance between bias and variance is a major challenge in machine learning.

1.6 Machine Learning Process

DATA MINING & MACHINE LEARNING FRAMEWORK

CRISP-DM Process Model

Cross-Industry Standard Process for Data Mining: A structured blueprint for end-to-end Machine Learning solutions.

6 Iterative Phases • Industry Standard

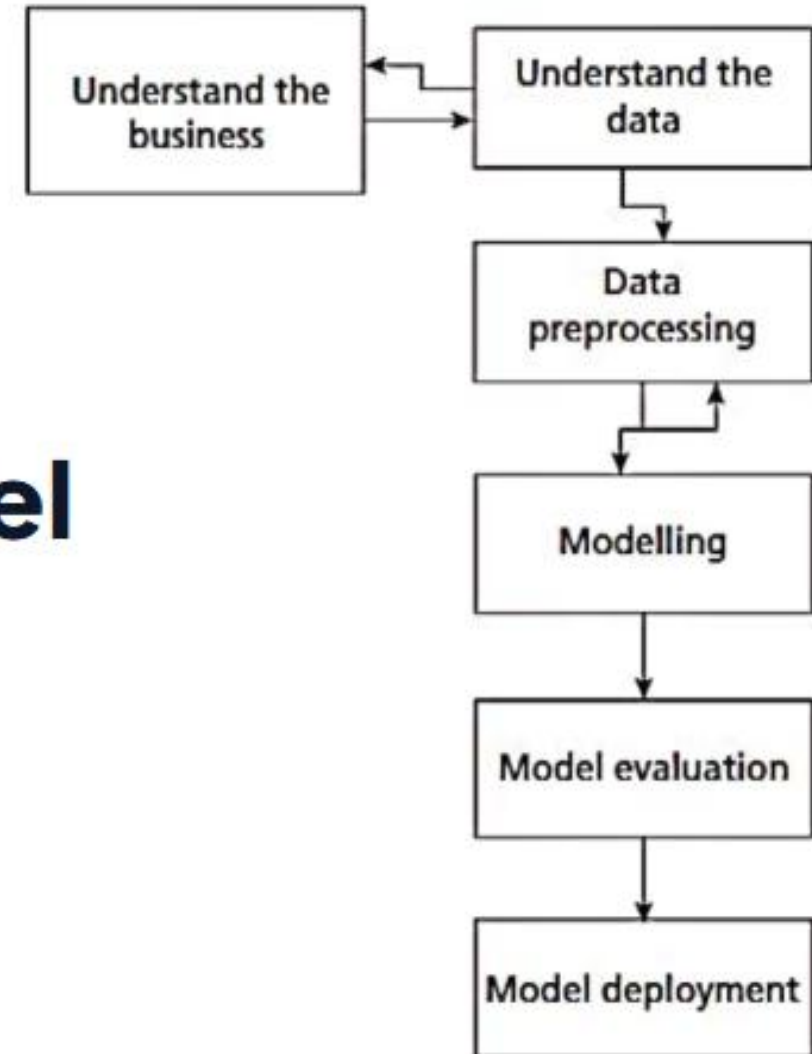


Figure 1.11: A Machine Learning/Data Mining Process

Understand the Business: Defines project objectives and requirements from a business perspective, converting them into a machine learning problem definition.

Understand the Data: Involves initial data collection, exploring data characteristics, identifying quality issues, and discovering first insights.

Data Preprocessing: Cleans raw data, handles missing values, normalizes features, and transforms datasets into the final format required for modeling.

Modelling: Selects and applies various machine learning algorithms to the prepared data and tunes model parameters for optimal performance.

Model Evaluation: Assesses the model against business metrics and key performance indicators (KPIs) to ensure it meets original business goals before deployment.

Model Deployment: Integrates the validated model into a live production environment so end-users or systems can make data-driven decisions.

1.7 Machine Learning Applications

Sentiment Analysis: An application of natural language processing (NLP) that converts words in documents into sentiments such as happy, sad, and angry, often captured by emoticons. It is used to automatically assign star ratings (e.g., five stars or one star) to movie or product reviews.

Recommendation Systems: Systems based on machine learning that enable personalized purchases. Examples include Amazon suggesting related books or books bought by people with similar tastes, and Netflix suggesting shows or movies based on user taste.

Voice Assistants: Products such as Amazon Alexa, Microsoft Cortana, Apple Siri, and Google Assistant that process speech commands to perform tasks.

Navigation & Mapping Technologies: Technologies such as Google Maps and Uber that locate and navigate the shortest paths to reduce travel time.

S.No.	Problem Domain	Applications
1.	Business	Predicting the bankruptcy of a business firm
2.	Banking	Prediction of bank loan defaulters and detecting credit card frauds
3.	Image Processing	Image search engines, object identification, image classification, and generating synthetic images
4.	Audio/Voice	Chatbots like Alexa, Microsoft Cortana. Developing chatbots for customer support, speech to text, and text to voice
5.	Telecommuni- cation	Trend analysis and identification of bogus calls, fraudulent calls and its callers, churn analysis
6.	Marketing	Retail sales analysis, market basket analysis, product performance analysis, market segmentation analysis, and study of travel patterns of customers for marketing tours
7.	Games	Game programs for Chess, GO, and Atari video games
8.	Natural Language Translation	Google Translate, Text summarization, and sentiment analysis
9.	Web Analysis and Services	Identification of access patterns, detection of e-mail spams, viruses, personalized web services, search engines like Google, detection of promotion of user websites, and finding loyalty of users after web page layout modification
10.	Medicine	Prediction of diseases, given disease symptoms as cancer or diabetes. Prediction of effectiveness of the treatment using patient history and Chatbots to interact with patients like IBM Watson uses machine learning technologies.
11.	Multimedia and Security	Face recognition/identification, biometric projects like identification of a person from a large image or video database, and applications involving multimedia retrieval
12.	Scientific Domain	Discovery of new galaxies, identification of groups of houses based on house type/geographical location, identification of earthquake epicenters, and identification of similar land use

Understanding Data – 1: Introduction

2.3 Big Data Analysis Framework

Difference Between Data Analytics and Data Analysis

- **Data Analytics:** A general, broader concept that covers the complete cycle of data management, including data collection, preprocessing, and analysis. It focuses more on the future and aids in making predictions.
- **Data Analysis:** A specific component within data analytics. It focuses on taking historical data and analyzing it to extract useful information and insights for decision-making.

4 Types of Data Analytics

1. Descriptive analytics
2. Diagnostic analytics
3. Predictive analytics
4. Prescriptive analytics

Descriptive Analytics: Focuses on describing and quantifying the main features of collected data. It relies strictly on the descriptive aspect of statistics rather than inference.

Example:

Monthly Sales Summary: A retail store generates end-of-month reports showing total revenue, units sold per location, and top-selling product lines.

Website Traffic Metrics: An e-commerce company tracks daily visitor counts, page views, and average session durations across different geographic regions using an analytics dashboard.

Diagnostic Analytics: Known as causal analysis, it answers the question "*Why?*" by determining the cause and effect of events. For instance, if a product is not selling, it aims to analyze the underlying reasons and associated effects.

Example:

Checkout Drop-off Investigation: An online retailer notices a sudden 20% drop in conversions and analyzes site log data to discover that a bug in the newly updated payment gateway was causing transaction failures.

Customer Churn Root-Cause Analysis: A subscription service analyzes user engagement patterns and customer support logs to determine that poor app performance after a software update led to increased cancellations.

Predictive Analytics: Deals with the future by answering "*What will happen in the future given this data?*" It applies algorithms to discover patterns and predict outcomes, forming the core focus of machine learning.

Example:

Seasonal Demand Forecasting: A supermarket chain uses historical sales trends, weather predictions, and upcoming holiday schedules to estimate how much inventory to stock for the next quarter.

Credit Default Risk Scoring: A bank uses machine learning models trained on past customer payment histories and credit scores to estimate the likelihood of a loan applicant defaulting.

Prescriptive Analytics: Recommends the best course of action for business organizations. Going beyond prediction, it suggests specific sets of actions to assist decision-making, improve future planning, and mitigate risks.

Example:

Dynamic Delivery Route Optimization: Logistics companies like UPS evaluate real-time traffic conditions, weather alerts, and delivery time windows to automatically prescribe the fastest, most fuel-efficient routes for drivers.

Automated Airline Price Adjustments: An airline uses algorithms that evaluate real-time booking speeds, competitor pricing, and remaining seat capacity to automatically adjust dynamic ticket prices to maximize profitability.

2.4 Descriptive Statistics

- Descriptive statistics is a branch of statistics used to summarize, organize, and describe data. It helps us understand the important characteristics of a dataset without making predictions.
- To understand the basic nature and characteristics of data.
- Represents data using graphs, charts, and plots.
- Examines data to discover patterns, relationships, and unusual observations.
- Understanding the data helps us decide which machine learning or data-mining techniques can be applied.
- Descriptive statistics + Data visualization are the main Components of EDA

Example: Suppose we have students' marks:

45, 60, 70, 75, 80, 90

Descriptive statistics can tell us:

- **Mean** → Average mark
- **Median** → Middle mark
- **Minimum** → 45
- **Maximum** → 90
- **Range** → $90 - 45 = 45$

A **bar chart or histogram** can visually show how the marks are distributed.

2.5 Univariate Data Analysis and Visualization

Univariate analysis is the simplest form of statistical analysis where the dataset consists of only one variable. It focuses purely on describing the data and identifying patterns within a single category, rather than dealing with causes or relationships between variables.

Key components involved in describing univariate data include:

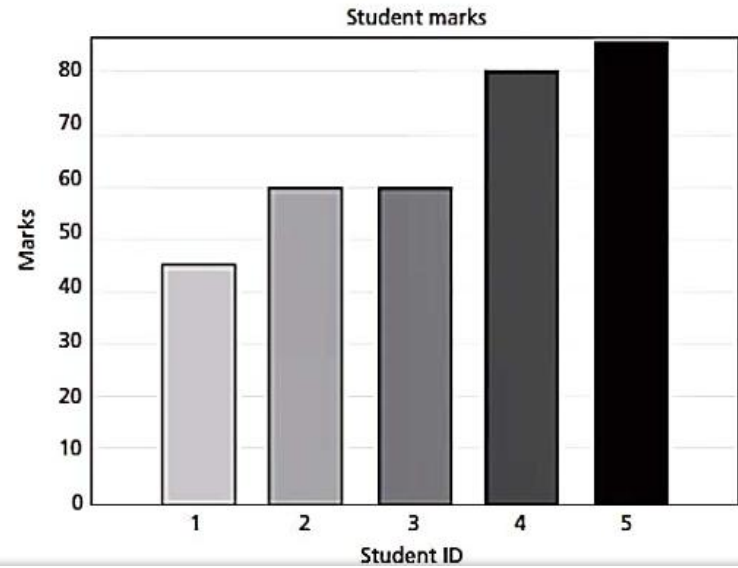
- Frequency Distributions
- Central Tendency Measures
- Dispersion or Variation
- Shape of the Data

To understand data, graph visualization is must. Data visualization helps to understand data. It helps to present information and data to customers. Some of the graphs that are used in univariate data analysis are bar charts, histograms, frequency polygons and pie charts.

2.5.1 Data Visualization

Bar Chart A Bar chart (or Bar graph) is used to display the frequency distribution for variables. Bar charts are used to illustrate discrete data. The charts can also help to explain the counts of nominal data. It also helps in comparing the frequency of different groups.

The bar chart for students' marks {45, 60, 60, 80, 85} with Student ID = {1, 2, 3, 4, 5} is shown below in Figure 2.3.



Histogram It plays an important role in data mining for showing frequency distributions. The histogram for students' marks {45, 60, 60, 80, 85} in the group range of 0–25, 26–50, 51–75, 76–100 is given below in Figure 2.5. One can visually inspect from Figure 2.5 that the number of students in the range 76–100 is 2.

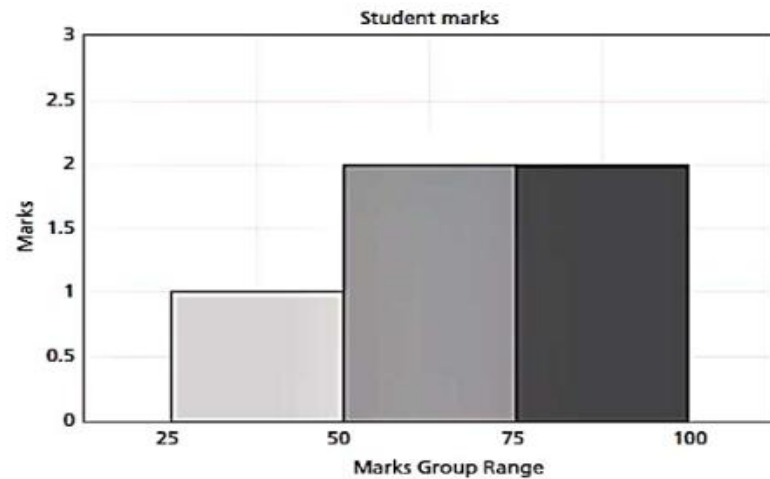


Figure 2.5: Sample Histogram of English Marks

Histogram conveys useful information like nature of data and its mode. Mode indicates the peak of dataset. In other words, histograms can be used as charts to show frequency, skewness present in the data, and shape.

Pie Chart These are equally helpful in illustrating the univariate data. The percentage frequency distribution of students' marks {22, 22, 40, 40, 70, 70, 70, 85, 90, 90} is below in Figure 2.4.

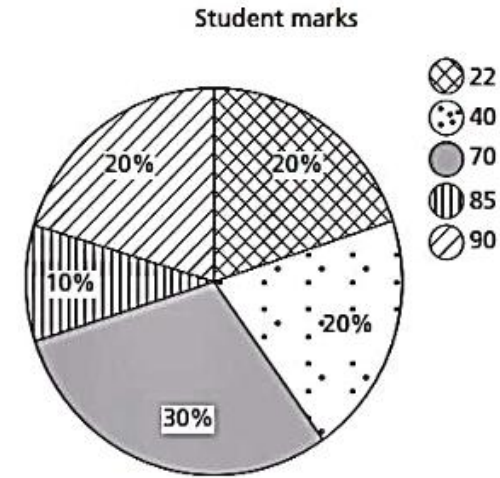


Figure 2.4: Pie Chart

It can be observed that the number of students with 22 marks are 2. The total number of students are 10. So, $2/10 \times 100 = 20\%$ space in a pie of 100% is allotted for marks 22 in Figure 2.4.

Dot Plots These are similar to bar charts. They are less clustered as compared to bar charts, as they illustrate the bars only with single points. The dot plot of English marks for five students with ID as {1, 2, 3, 4, 5} and marks {45, 60, 60, 80, 85} is given in Figure 2.6. The advantage is that by visual inspection one can find out who got more marks.

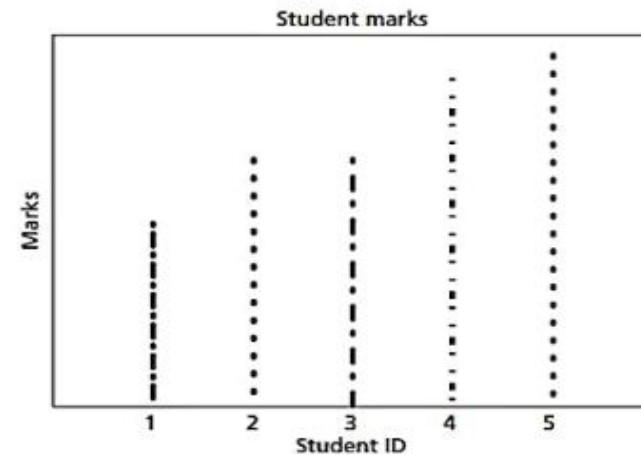


Figure 2.6: Dot Plots

2.5.2 Central Tendency

Central Tendency Overview: Condensing or summarizing data makes analysis and comparisons easier. Mass data tends to concentrate near a central location, known as a measure of central tendency or averages (the "first order of measures"). The most popular measures are mean, median, and mode.

Arithmetic Mean:

- Represents the 'center' of the dataset and is commonly used in everyday contexts (e.g., average income, average traffic).
- Calculated by adding all data values and dividing the sum by the total number of observations (N).
- Formula:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i$$

If you have N observations $x_1, x_2, \dots, x_N$ with corresponding weights $w_1, w_2, \dots, w_N$, the weighted mean ($\bar{x}_w$) is calculated as:

$$\bar{x}_w = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_N x_N}{w_1 + w_2 + \cdots + w_N}$$

Where:

- x_i = the value of each observation
- w_i = the weight or importance assigned to each observation

The **Geometric Mean** is defined as the N^{th} root of the product of N items.

$$\text{Geometric Mean} = \left(\prod_{i=1}^N x_i \right)^{\frac{1}{N}} = \sqrt[N]{x_1 \times x_2 \times \cdots \times x_N}$$

Where:

- N (or n) = total number of observations/items
- x_i = individual data values

Logarithmic / Anti-log Formula (for larger datasets): Because multiplying many numbers can be difficult, the geometric mean is commonly calculated using logarithms:

$$\text{Geometric Mean} = \text{anti-log} \left(\frac{\log(x_1) + \log(x_2) + \cdots + \log(x_N)}{N} \right) = \text{anti-log} \left(\frac{\sum_{i=1}^n \log(x_i)}{N} \right)$$

The **Median** is the middle value in a distribution, dividing the dataset into two equal halves where half the values are lower and half are higher.

Basic Rules (Ungrouped Data)

- **Odd number of items (N is odd):** The median is the exact middle value.
- **Even number of items (N is even):** The median is the average (mean) of the two central values.

Continuous / Grouped Data Formula

For continuous data grouped into classes, the median is calculated using:

$$\text{Median} = L_1 + \frac{\frac{N}{2} - cf}{f} \times i$$

Where:

- **Median class:** The class interval where the $(N/2)^{\text{th}}$ item lies.
- L_1 = Lower limit of the median class
- N = Total number of observations (total frequency)
- cf = Cumulative frequency of the class preceding the median class
- f = Frequency of the median class
- i = Class interval (width) of the median class

Mode

- It is the value that occurs most frequently in a dataset (the value with the highest frequency).
- It applies strictly to discrete data and is not used for continuous data since continuous data lacks repeated values.
- Calculate the frequencies of all individual values; the value (or values) with the highest frequency is the mode.

2.5.3 Dispersion

- Dispersion is the spread of a set of data around its central tendency (mean, median, or mode).
- Measures of dispersion are second-order measures. Common representations include range, variance, standard deviation, and standard error.
- **Range:** The difference between the maximum and minimum values in a given set of data.
- **Standard Deviation:** The average distance from the mean of the dataset to each data point. It helps measure the spread of data, which the mean alone does not capture (e.g., 10, 20, 30 and 10, 50, 0 both have a mean of 20, but different spread).

N represents the size of the population, X_i is an observation from the population, and mean is the population mean.

$N - 1$ is often used instead of N in the denominator because, for larger real-world datasets, division by $N - 1$ gives an answer closer to the actual value.

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}}$$

- **Percentiles & Quartiles:**
 - **Percentile:** The k^{th} percentile is the value where $k\%$ of the data lies at or below it (X_r).
 - **Median ($Q_{0.50}$):** The 50th percentile.
 - **First Quartile (Q_1):** The 25th percentile.
 - **Third Quartile (Q_3):** The 75th percentile.
- **Interquartile Range (IQR):**
 - Measures dispersion and is defined as the difference between Q_3 and Q_1 ($Q_{0.75} - Q_{0.25}$).
 - Formula: Interquartile percentile = $Q_3 - Q_1$.
- **Outliers:**
 - Defined as values falling apart by at least $1.5 \times \text{IQR}$ above the third quartile (Q_3) or below the first quartile (Q_1).

Example 2.4: For patients' age list {12, 14, 19, 22, 24, 26, 28, 31, 34}, find the IQR.

Solution: The median is in the fifth position. In this case, 24 is the median. The first quartile is median of the scores below the mean i.e., {12, 14, 19, 22}. Hence, it's the median of the list below 24. In this case, the median is the average of the second and third values, that is, $Q_{0.25} = 16.5$. Similarly, the third quartile is the median of the values above the median, that is {26, 28, 31, 34}. So, $Q_{0.75}$ is the average of the seventh and eighth score. In this case, it is $28 + 31/2 = 59/2 = 29.5$.

Hence, the IQR using Eq. (2.10) is:

$$\begin{aligned} &= Q_{0.75} - Q_{0.25} \\ &= 29.5 - 16.5 = 13 \end{aligned}$$

The half of IQR is called semi-quartile range. The Semi Inter Quartile Range (SIQR) is given as:

$$\begin{aligned} \text{SIQR} &= \frac{1}{2} \times \text{IQR} \\ &= \frac{1}{2} \times 13 = 6.5 \end{aligned}$$

Example 2.5: Find the 5-point summary of the list {13, 11, 2, 3, 4, 8, 9}.

Solution: The minimum is 2 and the maximum is 13. The Q_1 , Q_2 and Q_3 are 3, 8 and 11, respectively. Hence, 5-point summary is {2, 3, 8, 11, 13}, that is, {minimum, Q_1 , median, Q_3 , maximum}.

Box plots are useful for describing 5-point summary. The Box plot for the set is given in Figure 2.7.

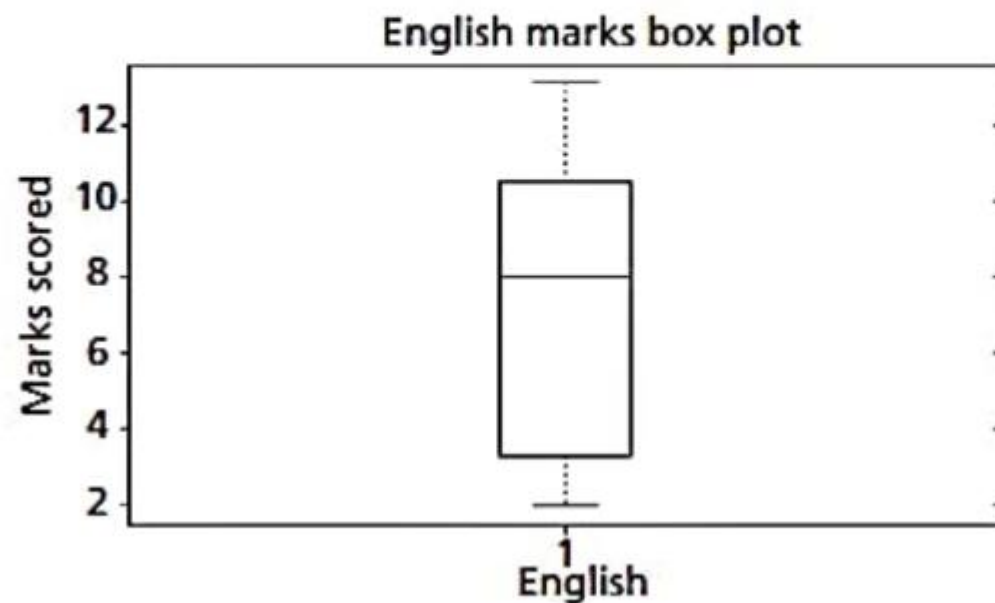


Figure 2.7: Box Plot for English Marks

2.5.4 Shape:

Skewness and Kurtosis (referred to as moments) indicate the symmetry/asymmetry and peak location of a dataset.

Skewness:

- **Order:** Measures of direction and degree of symmetry are called third-order measures.
- **Ideal Value:** In an ideal normal distribution, skewness should be zero.
- **Symmetry in Datasets:** More often than not, a given dataset does not display perfect symmetry.

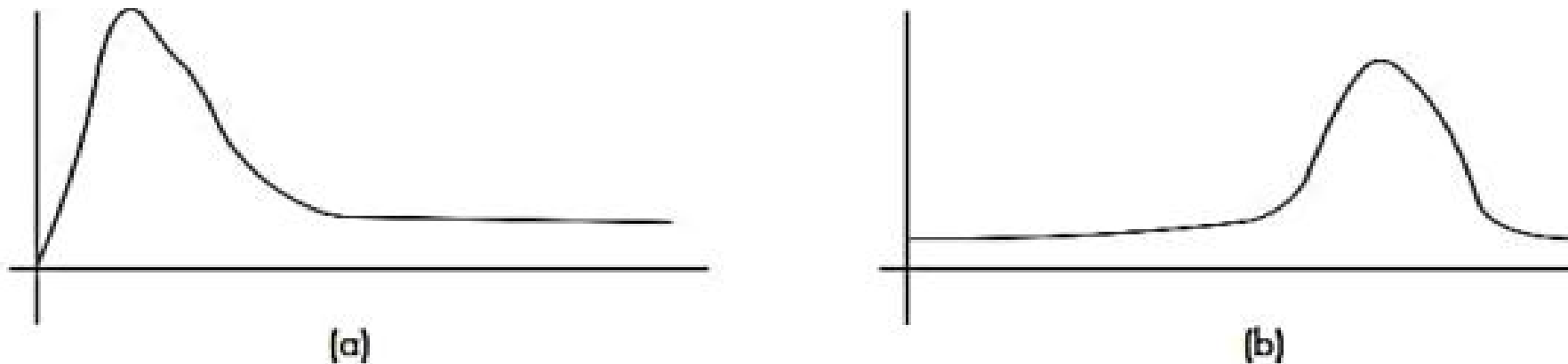


Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

- **Positively Skewed (Right-Skewed):**
 - Long tail extends to the **right** (positive side).
 - Mass/hump is on the **left**.
 - $\text{Mean} > \text{Median} > \text{Mode}$ (generally).
 - Pearson's coefficient is **positive** (> 0).
- **Negatively Skewed (Left-Skewed):**
 - Long tail extends to the **left** (negative side).
 - Mass/hump is on the **right**.
 - $\text{Mean} < \text{Median} < \text{Mode}$ (generally).
 - Pearson's coefficient is **negative** (< 0).

Kurtosis

Kurtosis describes the shape of a distribution by indicating the heaviness of its tails and the likelihood of extreme observations compared with a normal distribution.

The population kurtosis is given by:

$$K = \frac{\mu_4}{\sigma^4}$$

where:

- $\mu_4 = E[(X - \mu)^4]$ is the **fourth central moment**
- σ is the standard deviation.

For a normal distribution:

$$K = 3$$

The **excess kurtosis** is:

$$K_e = K - 3$$

Therefore:

- $K_e = 0 \rightarrow$ **Mesokurtic** (normal distribution)
- $K_e > 0 \rightarrow$ **Leptokurtic** (heavier tails)
- $K_e < 0 \rightarrow$ **Platykurtic** (lighter tails)

Mean Absolute Deviation (MAD)

Mean Absolute Deviation measures the average distance of observations from the mean. It indicates how much the observations vary around the mean.

The formula is:

$$MAD = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

where:

- x_i = individual observation
- $\bar{x}$ = mean of the observations
- n = number of observations
- $|x_i - \bar{x}|$ = absolute deviation from the mean.

For example, if the observations are:

10, 12, 14, 16, 18

Mean:

$$\bar{x} = 14$$

Absolute deviations:

4, 2, 0, 2, 4

Therefore:

$$MAD = \frac{4 + 2 + 0 + 2 + 4}{5} = 2.4$$

So, on average, the observations are **2.4 units away from the mean**.

Coefficient of Variation (CV)

Coefficient of Variation measures the relative variability of a dataset with respect to its mean. It is particularly useful for comparing the variability of datasets having different units or different means.

The formula is:

$$CV = \frac{\sigma}{\mu} \times 100\%$$

For a sample:

$$CV = \frac{s}{\bar{x}} \times 100\%$$

where:

- σ = population standard deviation
- μ = population mean
- s = sample standard deviation
- $\bar{x}$ = sample mean.

For example, if:

$$\bar{x} = 50, s = 5$$

then:

$$CV = \frac{5}{50} \times 100 = 10\%$$

A **lower CV** indicates greater consistency, while a **higher CV** indicates greater relative variability.

2.5.5 Special Univariate Plots

Univariate plots are graphical methods used to study **one variable at a time**. In addition to common plots such as histograms and box plots, several **specialized plots** are useful for understanding the distribution, spread, skewness, and unusual observations of a single variable.

1. Stem-and-Leaf

Plot

- Displays individual numerical observations while preserving their original values.
- Useful for small or moderately sized datasets.
- Helps identify the **center, spread, clusters, and outliers**.

Example:

Data = 12, 13, 15, 16, 21, 22, 25

Stem-and-Leaf Plot

1 | 2 3 5 6

2 | 1 2 5

Q–Q Plot (Quantile–Quantile Plot)

A Q–Q plot is a special univariate plot used to determine whether a dataset follows a particular theoretical distribution, most commonly the normal distribution.

A Q–Q plot compares:

- **X-axis:** Theoretical quantiles expected from a normal distribution
- **Y-axis:** Observed/sample quantiles obtained from the dataset

Example

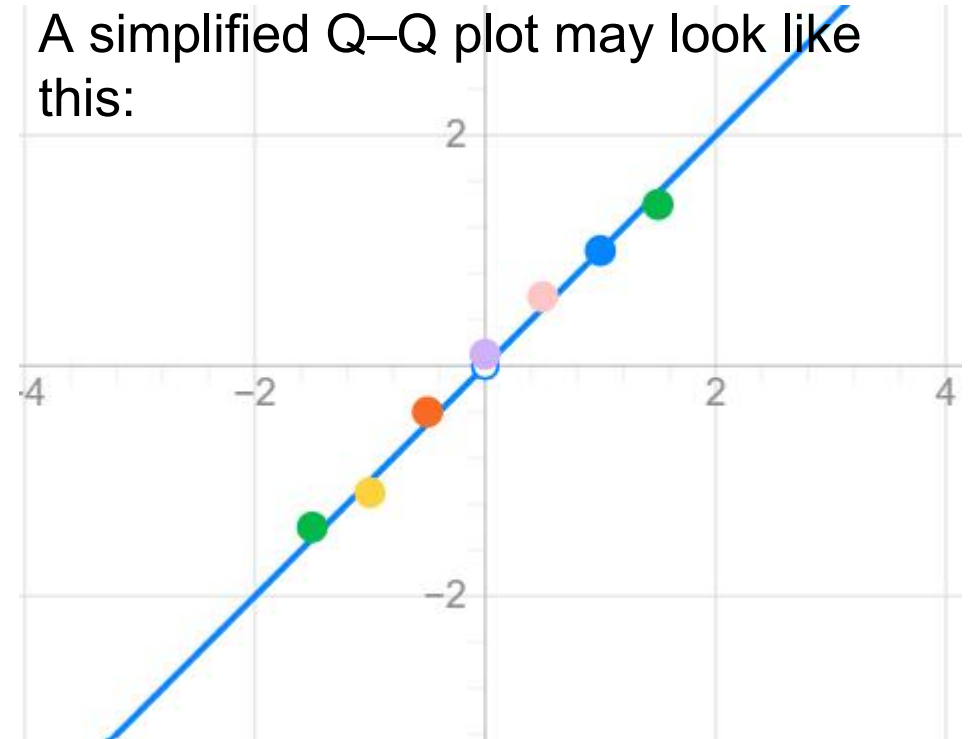
Suppose the observed data is:

10, 12, 13, 15, 16, 18, 20, 21, 23, 25

The observations are first arranged in ascending order and compared with the corresponding theoretical normal quantiles.

$y=x$
(-1.5,-1.4)
(-1,-1.1)
(-0.5,-0.4)
(0,0.1)
(0.5,0.6)
(1,1.0)
(1.5,1.4)

A simplified Q–Q plot may look like this:



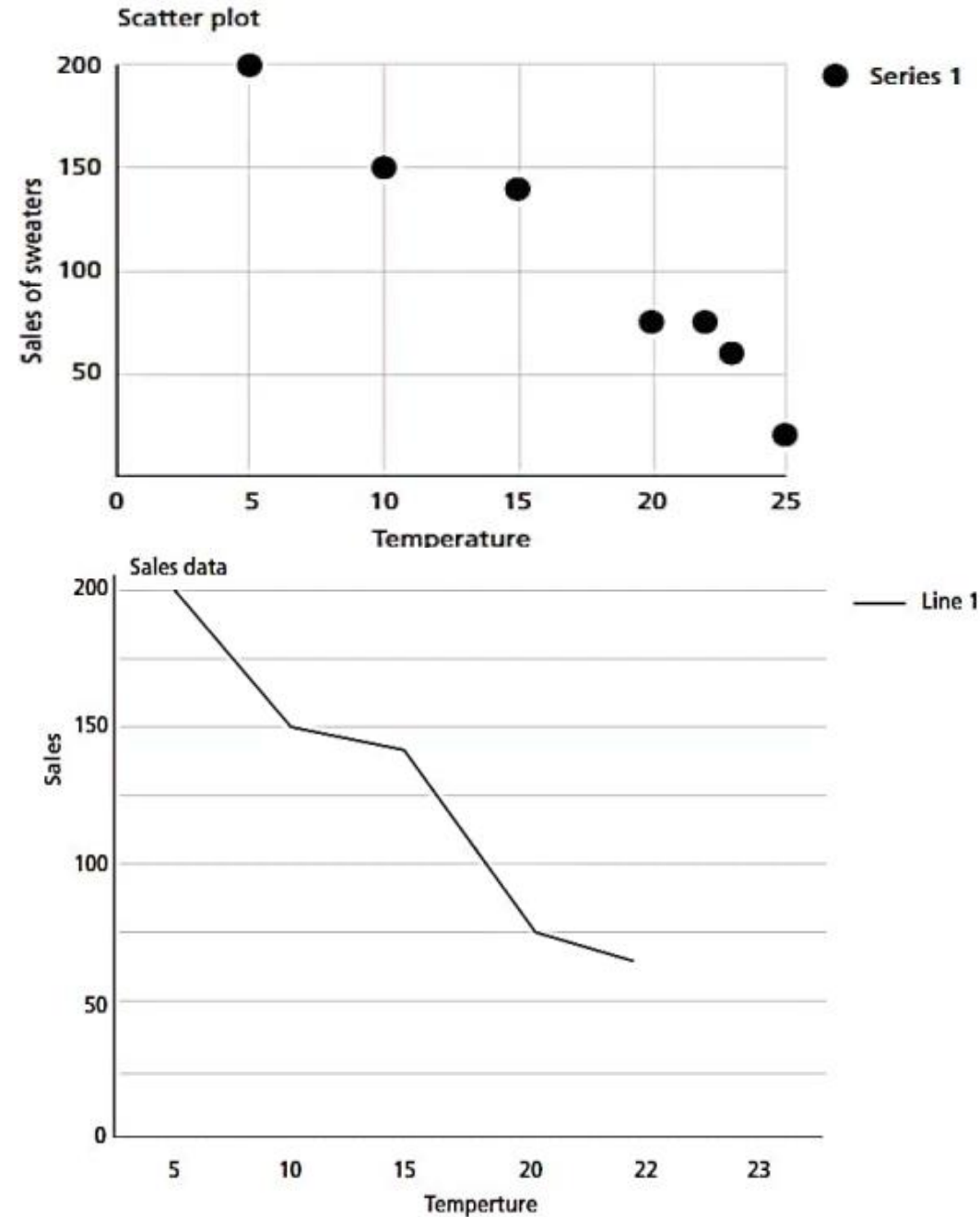
2.6 Bivariate Data and Multivariate Data

Bivariate data is a type of data that consists of two variables measured for the same set of observations. It is mainly used to study the relationship or association between the two variables.

Example:

- a. House area → House price
- b. Study Hours → Marks

A **scatter plot** and **Line graph** is a graphical method used to visualize bivariate data. Each observation is represented by a point:



Multivariate Data

Multivariate data consists of three or more variables measured or observed for the same set of individuals, objects, or observations.

It is used when we want to study the relationship between multiple variables simultaneously.

Example:

Suppose information about students is collected:

Here we have four variables:

- 1. Hours Studied
- 2. Attendance
- 3. Assignment Score
- 4. Examination Marks

Therefore, this is multivariate data.

Student	Hours Studied	Attendance (%)	Assignment Score	Marks
A	2	70	60	45
B	3	75	65	50
C	4	80	72	60
D	5	90	85	68
E	6	95	90	75

Feature	Bivariate Data	Multivariate Data
Meaning	Data involving two variables	Data involving three or more variables
Number of variables	Exactly 2	3 or more
Main purpose	Study the relationship between two variables	Study relationships among multiple variables
Example	Height and weight	Height, weight, age, and blood pressure
Typical analysis	Correlation, covariance, simple linear regression	Multiple regression, PCA, clustering, multivariate analysis
Common visualization	Scatter plot	Pair plot, heatmap, 3D scatter plot, parallel coordinates
Example in ML	Hours studied → Marks	Hours studied + Attendance + Previous marks → Final marks

Bivariate statistics: Covariance

Covariance and correlation are examples of bivariate statistics. Covariance is a measure of the joint probability of random variables (such as X and Y) and is used to measure the variance between two dimensions.

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(X))(y_i - E(Y))$$

- x_i and y_i are data values from X and Y .
- $E(X)$ and $E(Y)$ are the mean values of x_i and y_i .
- N is the total number of given data points.

Interpretation of Covariance

Covariance	Interpretation
Positive	X and Y tend to increase together
Negative	When X increases, Y tends to decrease
Approximately 0	No clear linear relationship between X and Y

Correlation:

Correlation is a statistical measure that quantifies the strength and direction of a linear relationship between two variables. While **covariance** indicates whether two variables change together, correlation standardizes this measure—scale-independent and bounded—making it easy to evaluate how strongly the variables are related.

1. Direction of Correlation

The direction describes how one variable changes relative to another:

- **Positive Correlation:** As one variable increases, the other variable tends to increase as well (e.g., study time and exam scores).
- **Negative Correlation:** As one variable increases, the other tends to decrease (e.g., speed of a car and travel time to a destination).
- **Zero Correlation:** Changes in one variable show no linear relation to changes in the other (e.g., shoe size and intelligence quotient).

2. The most common measure of linear correlation is the **Pearson correlation coefficient** (r). It normalizes the covariance of two variables by the product of their standard deviations.

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}$$

3. Interpreting the Values of r

The value of r always falls strictly within the range $-1 \leq r \leq 1$:

- $r = +1$: Perfect positive linear relationship.
- $0.7 \leq r < 1$: Strong positive linear relationship.
- $0 < r < 0.3$: Weak positive linear relationship.
- $r = 0$: No linear relationship.
- $-0.3 < r < 0$: Weak negative linear relationship.
- $-1 < r \leq -0.7$: Strong negative linear relationship.
- $r = -1$: Perfect negative linear relationship.

Where:

- $\text{cov}(X, Y)$ is the covariance between X and Y .
- σ_X and σ_Y are the standard deviations of X and Y .
- $\bar{x}$ and $\bar{y}$ are the sample means of X and Y .

4. A foundational rule in statistics is that **correlation does not imply causation**:

- High correlation simply means two variables move together predictably.
- It does not mean that changes in one variable cause changes in the other; the relationship may be coincidental or driven by a third, unseen variable (a confounding factor).

Solved exercises on correlation and covariance

2.7 Multivariate Statistics

Multivariate statistics is a branch of statistics that deals with the simultaneous analysis of two or more variables for each observation. It helps us understand the relationships, patterns, dependencies, and combined effects among multiple variables.

Example:

We have four variables:

- Study Hours
- Attendance
- Assignment Score
- Exam Score

Since more than two variables are analyzed together, this is **multivariate data**, and statistical methods used to analyze it are called **multivariate statistics**.

Heatmap

Display correlations or values using intensity

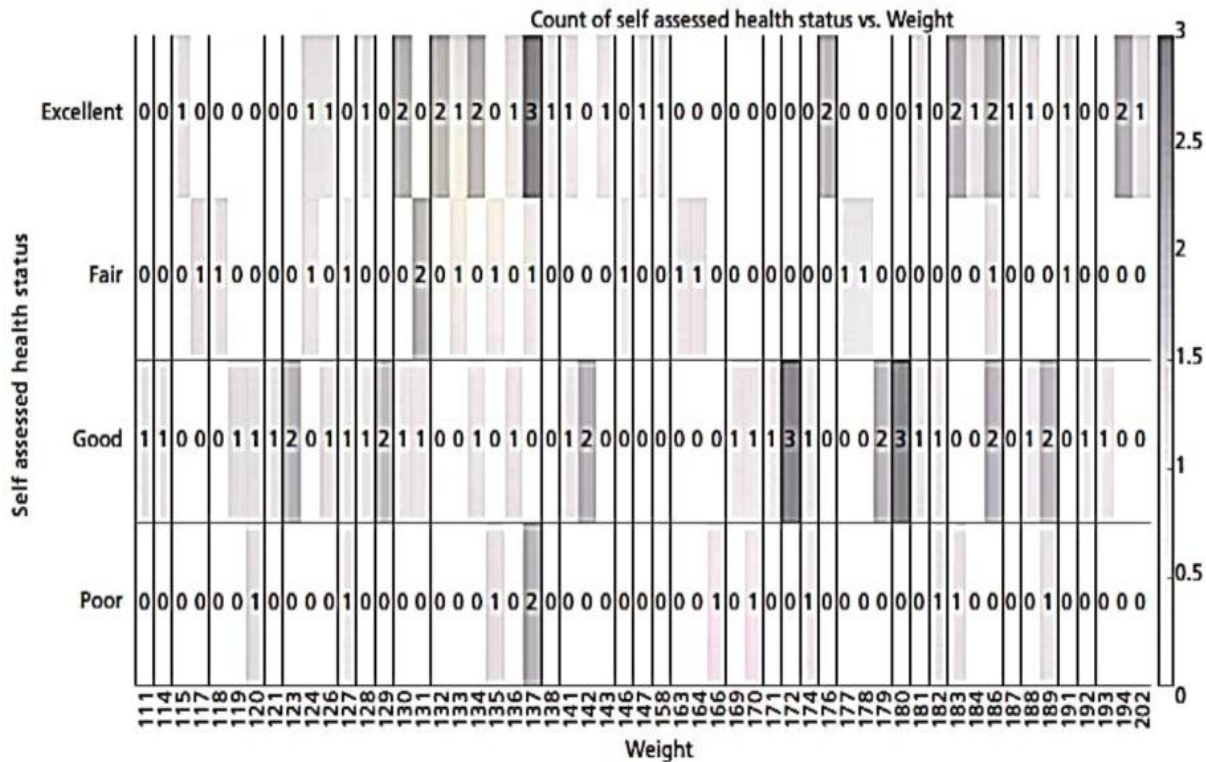


Figure 2.13: Heatmap for Patient Data

Pair Plot

Combine scatter plots and distributions

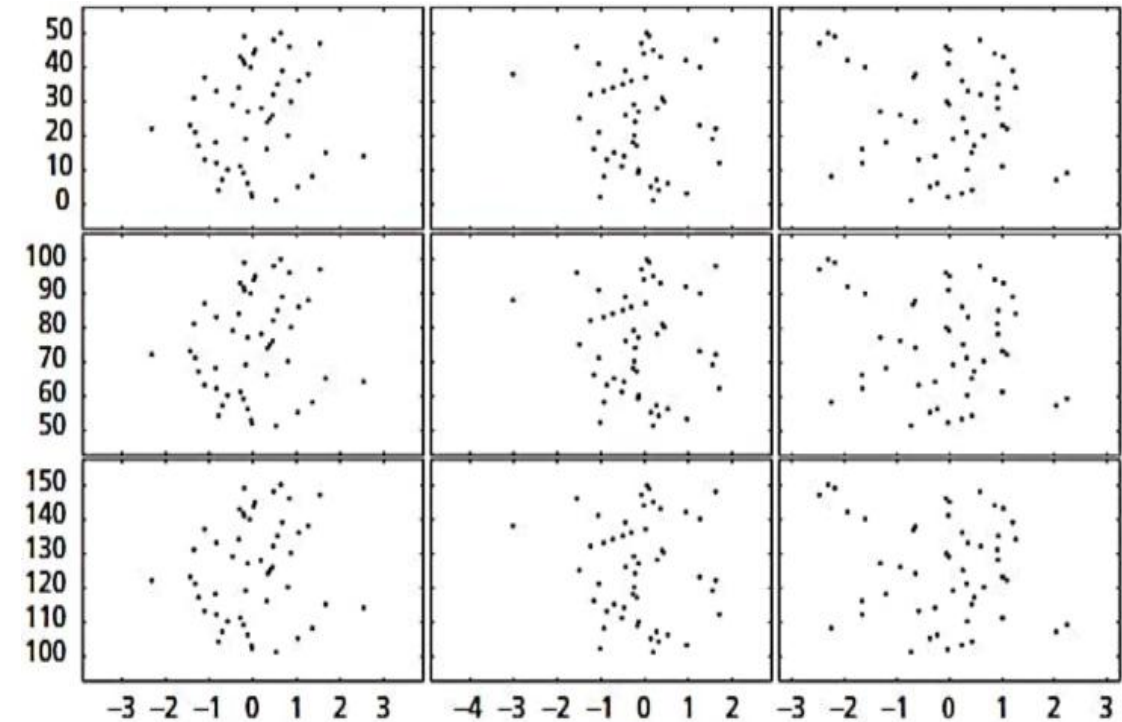


Figure 2.14: Pairplot for Random Data

END OF UNIT-1